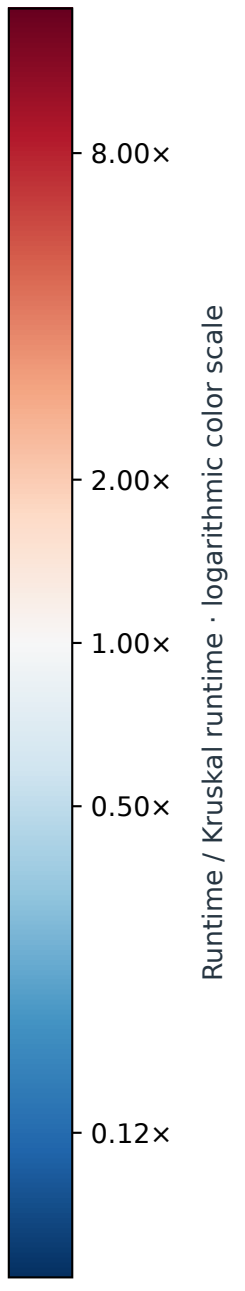
MST runtime relative to Kruskal

Lower is faster · blue < 1× · red > 1× · all measured cases retained

	KKT	Kruskal	Prim	Borůvka
Random · m = 4n · n = 1,024	1.84×	1.00×	2.91×	1.41×
Random · m = 4n · n = 4,096	2.07×	1.00×	2.93×	1.50×
Random · m = 4n · n = 16,384	2.43×	1.00×	3.47×	1.66×
Random · m = 4n · n = 65,536	2.62×	1.00×	3.96×	1.82×
Random · m = 32n · n = 1,024	0.58×	1.00×	0.76×	0.33×
Random · m = 32n · n = 4,096	0.87×	1.00×	0.96×	0.33×
Random · m = 32n · n = 16,384	1.52×	1.00×	1.13×	0.39×
Random · m = 32n · n = 65,536	4.19×	1.00×	1.34×	0.53×
Random · m = 128n · n = 1,024	0.37×	1.00×	0.44×	0.18×
Random · m = 128n · n = 4,096	0.54×	1.00×	0.50×	0.18×
Random · m = 128n · n = 16,384	1.42×	1.00×	0.55×	0.25×
Dense random anchor · n = 1,000	0.29×	1.00×	0.29×	0.19×
Grid · n = 1,024	2.60×	1.00×	3.78×	2.91×
Grid · n = 4,096	1.81×	1.00×	2.87×	2.62×
Grid · n = 16,384	1.75×	1.00×	2.95×	2.93×
Geometric grid · n = 1,024	1.62×	1.00×	2.90×	1.61×
Geometric grid · n = 4,096	1.22×	1.00×	2.45×	1.55×
Geometric grid · n = 16,384	1.20×	1.00×	2.55×	1.90×
Hub · n = 1,024	1.07×	1.00×	3.19×	1.26×
Hub · n = 4,096	0.90×	1.00×	2.93×	1.13×
Hub · n = 16,384	0.91×	1.00×	3.34×	1.20×
Clustered · n = 1,024	0.48×	1.00×	1.03×	0.47×
Clustered · n = 4,096	0.63×	1.00×	1.15×	0.52×
Clustered · n = 16,384	0.82×	1.00×	1.26×	0.56×
Planted tree · n = 1,024	2.07×	1.00×	3.34×	1.54×
Planted tree · n = 4,096	1.99×	1.00×	2.92×	1.39×
Planted tree · n = 16,384	2.31×	1.00×	3.36×	1.54×
Random tree · n = 1,024	4.24×	1.00×	5.73×	6.98×
Random tree · n = 4,096	2.39×	1.00×	3.46×	4.61×
Random tree · n = 16,384	1.93×	1.00×	2.93×	4.58×
Star · n = 1,024	2.09×	1.00×	4.70×	4.04×
Star · n = 4,096	1.53×	1.00×	4.65×	2.94×
Star · n = 16,384	0.93×	1.00×	3.61×	1.69×
Hierarchical path · n = 1,024	14.8×	1.00×	5.66×	12.9×
Hierarchical path · n = 4,096	8.00×	1.00×	2.94×	11.5×
Hierarchical path · n = 16,384	5.52×	1.00×	1.95×	9.82×
Complete graph · n = 256	0.28×	1.00×	0.37×	0.20×
Complete graph · n = 512	0.26×	1.00×	0.30×	0.16×
Complete graph · n = 1,024	0.30×	1.00×	0.26×	0.12×



Paired ratios: median over repetitions → sampling seeds → graph seeds. No confidence intervals.